

BLM2041 Signals and Systems

The Instructors:

Prof. Dr. Nizamettin Aydın

naydin@yildiz.edu.tr

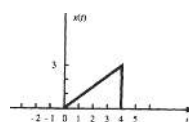
Asist. Prof. Dr. Ferkan Yilmaz

ferkan@yildiz.edu.tr

1

Example 1

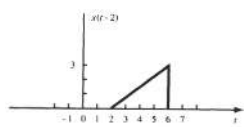
- Given the following continuous-time signal $x(t)$;



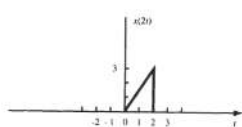
- sketch and label each of the following signals.
- (a) $x(t - 2)$;
- (b) $x(2t)$;
- (c) $x(t/2)$;
- (d) $x(-t)$

2

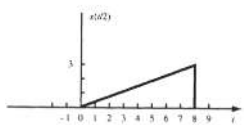
Answer 1



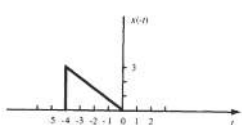
(a)



(b)



(c)

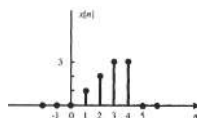


(d)

3

Example 2

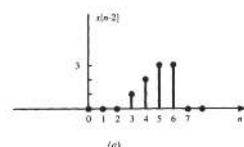
- Given the following discrete-time signal $x[n]$;



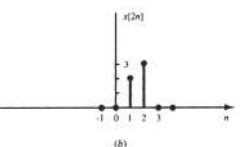
- (a) $x[n - 2]$;
- (b) $x[2n]$;
- (c) $x[-n]$;
- (d) $x[-n + 2]$

4

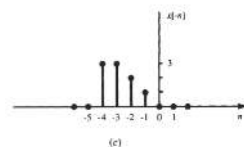
Answer 2



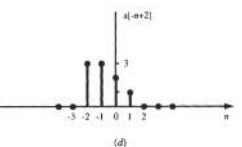
(a)



(b)



(c)

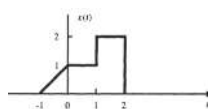


(d)

5

Example 3

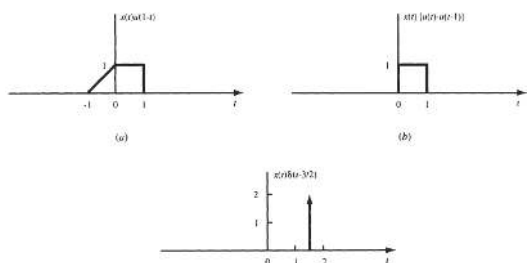
- Given the following continuous-time signal $x(t)$;



- sketch and label each of the following signals.
- (a) $x(t)u(1-t)$;
- (b) $x(t)[u(t) - u(t-1)]$;
- (c) $x(t/2)\delta(t-1.5)$;

6

Answer 3



7

Example 4

- Find the energy content of the exponentially decreasing signal $x(t)$

$$x(t) = \begin{cases} e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

8

Answer 4

- 1st, compute the square
 $|x(t)|^2 = (e^{-2t})^2 = e^{-4t}$
- Considering that the signal is zero for $t < 0$,

$$E = \int_0^{\infty} |x(t)|^2 dt = \int_0^{\infty} e^{-4t} dt = -\frac{1}{4} e^{-4t} \Big|_0^{\infty}$$

$$E = -\frac{1}{4} e^{-4t} \Big|_0^{\infty} = \left[-\frac{1}{4} e^{-4 \times \infty} + \frac{1}{4} e^{-4 \times 0} \right]$$

$$E = \left[-\frac{1}{4} \times 0 + \frac{1}{4} \times 1 \right] = \frac{1}{4}$$
- The energy is finite,
 – so this is an energy signal.

9

Example 5

- Let $x(t) = A \cos \omega t$, where A is a positive real constant.
- Find
 - (a) the signal energy over one period
 - (b) the average power of the signal

10

Answer 5

- (a) The period of this signal: $T_0 = \frac{2\pi}{\omega}$
- Square of signal: $\cos^2 x = \frac{1 + \cos 2x}{2}$
- The energy over one period is
- $$E_0 = \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt = \int_{-T_0/2}^{T_0/2} |A \cos \omega t|^2 dt = A^2 \int_{-T_0/2}^{T_0/2} \cos^2 \omega t dt$$
- $$E_0 = A^2 \int_{-T_0/2}^{T_0/2} \frac{1 + \cos 2\omega t}{2} dt = \frac{A^2}{2} \int_{-T_0/2}^{T_0/2} dt + \frac{A^2}{2} \int_{-T_0/2}^{T_0/2} \cos 2\omega t dt$$

11

Answer 5

- $$\frac{A^2}{2} \int_{-T_0/2}^{T_0/2} dt = \frac{A^2}{2} t \Big|_{-T_0/2}^{T_0/2} = \frac{A^2}{2} \left(\frac{T_0}{2} + \frac{T_0}{2} \right) = \frac{A^2}{2} T_0$$
- $$\int_{-T_0/2}^{T_0/2} \cos 2\omega t dt = \frac{1}{2\omega} \sin 2\omega t \Big|_{-T_0/2}^{T_0/2}$$
- $$= \frac{1}{2\omega} [\sin(\omega T_0) - \sin(-\omega T_0)] = \frac{\sin(\omega T_0)}{\omega}$$
- $$\int_{-T_0/2}^{T_0/2} \cos 2\omega t dt = \frac{\sin(\omega T_0)}{\omega} = \frac{\sin(2\pi)}{\omega} = 0 \quad E_0 = \frac{A^2}{2} T_0$$
- (b) Average power:
- $$P = \frac{E_0}{T_0} = \frac{A^2 T_0 / 2}{T_0} = \frac{A^2}{2}$$

12

Example 6

- Consider a signal $x(t) = e^{-|t|}$.
Determine the energy and power content of this signal.

13

Answer 6

- Compute the squared modulus of the function

$$|x(t)|^2 = e^{-2|t|} \quad |x(t)|^2 = \begin{cases} e^{2t} & \text{for } t < 0 \\ e^{-2t} & \text{for } t > 0 \end{cases}$$

- Split the integral into two parts, and perform the calculation

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^0 e^{2t} dt + \int_0^{\infty} e^{-2t} dt = 2 \int_0^{\infty} e^{-2t} dt = 1$$

- The energy is finite (energy signal)

14

Answer 6

- To find average power, compute

$$\frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \frac{1}{T} \int_{-T/2}^{T/2} e^{-2|t|} dt = \frac{2}{T} \int_0^{T/2} e^{-2t} dt = \frac{1}{T} (1 - e^{-T})$$

- Take the limit:

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} (1 - e^{-T}) = \lim_{T \rightarrow \infty} \frac{1}{T} - \lim_{T \rightarrow \infty} \frac{e^{-T}}{T}$$

– The first term vanishes.

– For the second term, notice that as $T \rightarrow \infty$, $e^{-T} \rightarrow 0$.

- Therefore the second term vanishes as well, and we have $P = 0$ as expected for an energy signal.

15

Example 7

- Find the even and odd components of
 $x(t) = 2\cos t - \sin t + 3\sin t \cos t$

- Reminder:

$$x_e(t) = \frac{x(t) + x(-t)}{2} \quad x_o(t) = \frac{x(t) - x(-t)}{2}$$

16

Answer 7

- 1st, find $x(-t)$

$$\begin{aligned} x(-t) &= 2\cos(-t) - \sin(-t) + 3\sin(-t)\cos(-t) \\ x(-t) &= 2\cos t + \sin t - 3\sin t \cos t \end{aligned}$$

- Even component

$$x_e(t) = \frac{x(t) + x(-t)}{2} = \frac{4 \cos t}{2} = 2 \cos t$$

- Odd component

$$x_o(t) = \frac{x(t) - x(-t)}{2} = \frac{-2 \sin t + 6 \sin t \cos t}{2} = -\sin t + 3 \sin t \cos t$$

17

18